Ethan Villalovoz

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Contact

Georgia Institute of Technology
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Links: Website, LinkedIn, Google Scholar

Research Interests

Robot learning and human–robot interaction, with interests in robot policy learning, reward learning, uncertainty-aware interaction, and aligning robot behavior with human intent under ambiguity.

Education

Georgia Institute of Technology — **Jan 2026 – May 2028**
M.S. in Computer Science, Computational Perception and Robotics — GPA: 4.0/4.0

Washington State University — **Aug 2021 – May 2025**
B.S. in Computer Science, Minor in Mathematics — GPA: 3.94/4.0

Preprints

[P1] **An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models**.
S. Tomar, A. Deshwal, E. Villalovoz, M. Fazzini, H. Cai, J.R. Doppa.
arXiv:2512.15076, 2025.

Conference Publications

[C1] **Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach**.
A. Bacula, E. Villalovoz, D. Flynn, A. Mehta, H. Knight.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.

Experience

Georgia Institute of Technology, Atlanta, Georgia, United States — **Feb 2026 – Present**
Graduate Student Researcher, Advised by Animesh Garg
Developing a video-to-action robot policy learning framework coupling WAN 2.2 video diffusion with action flow matching; building PyTorch/Hydra pipelines for LIBERO rollouts and ablations.

Microsoft, Redmond, Washington, United States — **May 2026 – Present**
Software Engineer Intern, Advised by Karunakar Vecham
Building a multi-agent workflow system that integrates agent orchestration into an internal cloud platform.

Washington State University, Pullman, Washington, United States — **Jan 2024 – May 2025**
Undergraduate Research Assistant, Advised by Janardhan Rao (Jana) Doppa
Implemented evaluation code and reproducible benchmarks for BODE-GEN, a Bayesian prompt-optimization method for test-driven LLM code generation across HumanEval+ tasks.

Carnegie Mellon University, Pittsburgh, Pennsylvania, United States — **Jun 2024 – Aug 2024**
Robotics Institute Summer Scholar, Advised by Henny Admoni
Developed a Bayesian reward-learning system for simulated household manipulation from state-correction and feature-clarification interactions.

Google, Sunnyvale, California, United States — **May 2023 – Aug 2023**
Software Engineering Intern (STEP), Advised by Arun Tej Chennadi, Paul Valdez
Developed 5 C++/SQL analytics jobs for internal database queue metrics and dashboard tools supporting engineering data workflows.

Oregon State University, Corvallis, Oregon, United States — **Jun 2022 – Aug 2022**
NSF REU Fellow, Advised by Heather Knight
Designed geometric motion primitives and Python/Tkinter tools for expressive formations in human-robot interaction research.

Awards & Honors

CS Research Mentorship Program Scholar, Google Research 2023

Selected for a structured mentorship program supporting undergraduate pathways into computer science research.

Generation Google Scholarship 2023

Awarded for academic performance, leadership, and commitment to diversity, equity, and inclusion in computing.

NIH MARC Scholar - National Institutes of Health (T34) 2023 – 2025

Selected for a two-year NIH-funded undergraduate research training program supporting scientific research, leadership development, and graduate-school preparation.

Teaching

CPT_S 315: Introduction to Data Mining Spring 2025

Undergraduate Teaching Assistant, Washington State University

CPT_S 350: Design and Analysis of Algorithms Fall 2024

Undergraduate Teaching Assistant, Washington State University

CPT_S 355: Programming Language Design Fall 2023

Undergraduate Teaching Assistant, Washington State University

CPT_S 121: Program Design and Development C/C++ Fall 2022

Undergraduate Teaching Assistant, Washington State University

Outreach

WSU MARC & MIRA Program (Invited Talk) 2025

Invited to speak with undergraduate researchers about graduate-school applications and strategies for pursuing research opportunities.

WSU VCEA (College Ambassador) 2022 – 2024

Represented Voiland College to industry partners, alumni, and prospective students.

CMU RISS RoboLaunch (Website Coordinator) 2024

Updated the RoboLaunch website for a robotics outreach initiative featuring talks and interactive workshops.